



Tillicum Slurm - IMT 526 A

Kristen Finch
Director of Solutions



INFORMATION TECHNOLOGY

UNIVERSITY *of* WASHINGTON

IMT526A - Group Projects

- Scaling up workflows
 - larger models, more data
 - **longer run time**
- Moving from interactive to unsupervised project workflows

Slurm - Simple Linux Utility Resource Management

Tillicum's system that:

- Allocates GPUs, CPUs, and memory
- Schedules when your jobs run
- Ensures fair sharing across all users

Using Slurm allows you to:

- Run longer jobs without staying connected
- Use GPUs efficiently
- Run multiple experiments for your project simultaneously

Scheduling Jobs

Job Types:

- Interactive - `salloc` for testing
- Batch - `sbatch` for project workloads

Quality of Service (QoS):

- normal - 16 GPUs for up to 24 hours - project workloads
- debug - 1 GPU for 1 hour - quick testing and setup
- interactive - 2 GPUs for 8 hours - short exploratory sessions

Group Slurm Account

Each project group has been assigned its own Slurm account:

- group1, group2, group3, etc.

Key Slurm flag:

--account=groupX

Group Slurm Account in Open OnDemand

Home / My Interactive Sessions / Jupyter

Interactive Apps

Desktops

Tillicum Desktop

IDEs

Jupyter

VS Code

Jupyter

This app will launch a Jupyter server on one or more nodes.

Account

✓ group1

imt526a

io500

uwit

debug

Number of GPUs

1

Number of hours

1

Number of minutes

0

User Interface

☒ Jupyter Lab

☐ Jupyter Notebook

☐ Modify Jupyter command-line options?

☐ Modify Slurm command-line options?

Launch

* The Jupyter session data for this session can be accessed under the [data root directory](#).

Monitoring Your Usage

- `hyakusage``: CLI tool to help users track their usage and costs
 - Displays usage for all accounts you have access to.
 - Shows total GPU-hours and costs per account for current billing cycle.
 - Shows active credits (e.g., remaining demo credits) with corresponding expiration dates.
 - Integrates user- and account-level budgets (if set) to show progress toward limits.

- * Billing cycle starts on the 26th of each month.
- * Usage is counted by job `END` date.
- * Costs are calculated by base rate (\$ `0.9/h`) x billable GPU hours, where billable GPU hours = raw GPU hours x QOS multiplier.
- * Costs are rounded to nearest cent.

Usage Report for Account `account1`:
Current Billing Cycle **TOTAL** Usage: `26.23` GPU hours, \$`23.61`
TOTAL Active Credits: \$`437.34`

- * \$`347.34` expiring on `2026-02-25`
- * \$`90.00` (no expiration)

Usage User Breakdown (2025-12-26 to 2026-01-02)		
USER	GPU HOURS (HRS)	JOBS
user1	3.69	16

Usage QOS Breakdown (2025-12-26 to 2026-01-02)			
QOS	(MULTIPLIER)	GPU HOURS (HRS)	COST (USD)
interactive	(x 1.0)	3.44	\$3.09
normal	(x 1.0)	0.26	\$0.23
TOTAL (selected users)		3.69	\$3.33

Budget Usage Summary				
USER	USED/BUDGET (USD)	USED %	TYPE	ENFORCE
user1	3.69 / 500.00	1%	total	true
account1	23.61 / 1000.00	2%	monthly	false

Group Storage Directory

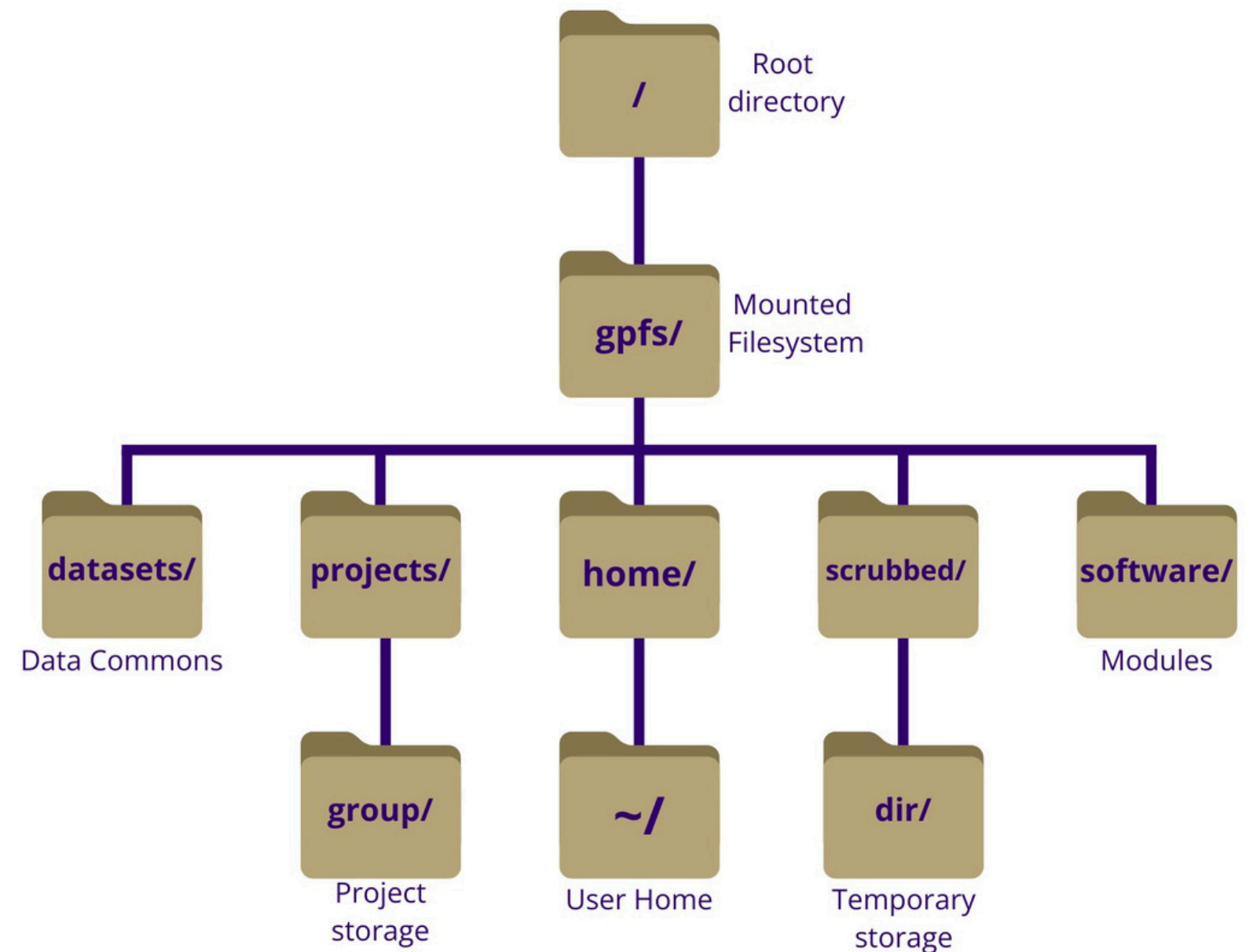
Each project group has been assigned its own group storage directory:

- /gpfs/projects/imt526a/group1
 - /gpfs/projects/imt526a/group2
 - /gpfs/projects/imt526a/group3
 - ...
-
- Store all inputs/outputs there
 - Avoid using home directories for project data

Filesystem and Storage on Tillicum

Storage spaces for Tillicum users:

- Home directory (/gpfs/home/UWNetID): **10GB**, personal, backed-up storage
- Project/lab dedicated storage (/gpfs/projects/imt526a): **20TB**, shared, backed-up storage for projects
- Scrubbed storage (/gpfs/scrubbed/\$USER): large, temporary scratch space for active computation



Typical Workflow

1. Develop and test code in Jupyter
2. Save your code as a Python script (or notebook)
3. Write a Slurm job script
4. Submit with `sbatch`
5. Check results in output files
6. Repeat

Hands-on

1. Clone the git repository into scrubbed for your practice
2. Convert the Python Demo Notebook into a Python Script
3. Start an interactive job
 - a. Test run the script
4. Submit the script to Slurm for unsupervised execution
5. View the results

<https://github.com/UWrc/tillicum-onboarding-imt526a>